Fiscal Year:	FY 2013	**Task Last Updated:**	FY 06/04/2013
PI Name:	Hargens, Alan R. Ph.D.		
Project Title:	Rodent spine deconditioning after 30 days of microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology (2) Neurobiology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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City:	La Jolla	**State:**	CA
Zip Code:	92037-0863	**Congressional District:**	52
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2007 Space Flight: Bion-M1 NNH07ZTT001N
Start Date:	08/01/2009	**End Date:**	07/31/2014
No. of Post Docs:	3	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
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Flight Program:	Bion/Shuttle		
Flight Assignment:	STS-131 ; Bion M1 NOTE: Extended to 7/31/2014 per A. Chu/ARC (Ed., 4/30/2014)		
Key Personnel Changes/Previous PI:	Jan 2012. New Co-I: Brandon R Macias, PhD, NSBRI Postdoctoral Fellow, Department of Orthopaedic Surgery, University of California, San Diego, San Diego, CA 92103-8894, Email: b1macias@ucsd.edu		
COI Name (Institution):	Lotz, Jeffrey (University of California, San Francisco) Masuda, Koichi (University of California, San Diego) Rana, Brinda (UCSD Cancer Center Sequencing Core) Sah, Robert (UCSD Bioengineering) Macias, Brandon (University of California, San Diego)		
Grant/Contract No.:	NNX09AP11G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The vertebral bodies and flexible intervertebral discs of the spine are important, weight-bearing tissues that are adapted to gravitational stress. The absence of gravitational axial loads on the body during exposure to microgravity may disrupt normal spine physiology and function. During microgravity exposure, deconditioning of the intervertebral discs and spinal tissues poses a serious risk upon re-exposure to 1-G and exercise. Our recent data in identical twins exposed to 30-days bed rest indicate that compressibility increases in the lumbar spine and its intervertebral discs during 30-days simulated microgravity. Because intervertebral discs become more collapsible during simulated microgravity, the probability of disc prolapse or herniation upon re-exposure to gravity increases. We hypothesize that nucleus pulposus swelling pressures in mouse spines will significantly decrease following 15- and 30-days of microgravity. In addition, we hypothesize that lack of gravitational loading on lumbar discs during microgravity will lead to extracellular matrix damage and significantly reduced disc compression performance following 15- and 30-days of microgravity. To test our hypotheses, we will harvest and test spines from ground-based control mice and flight mice exposed to 15 and 30-days of actual microgravity on STS-131 and Bion-M1. Spines will be frozen and transported to UCSD and UCSF for accepted methods of testing nucleus pulposus swelling pressure, lumbar disc dimensions, proteoglycan content, and spine biomechanics. We will compare the results of the present study to our peer-reviewed and published results for identical twins exposed to 30-days simulated microgravity and control mice exposed to 1-G. Our research will aid understanding of spinal deconditioning during 15- and 30-days microgravity and of the higher incidence of disc prolapse or herniation following re-exposure to 1-G with a long-term view to prevent such spinal deconditioning with exercise or other physiologic countermeasures. Separately, evidence suggests headward fluid shifts alone influence bone alteration. However, most conclusions are drawn from human bed rest studies and hindlimb unloading of rats aimed to simulate actual weightlessness. As a member of NASA's Biospecimen Sharing Program, our lab acquired murine calvariae from a 15-day mission aboard Space Shuttle STS-131. In concordance with current theory, we hypothesized that compared to ground control mice, headward fluid shifts in mice exposed to microgravity will induce adaptive growth in bones that normally do not bear weight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	To our knowledge, this study is the first to examine the effects of 15- and 30-days space flight on spinal deconditioning in mice. The vertebral bodies and flexible intervertebral discs are important, weight-bearing tissues that have adapted to gravitational stress. Our research will aid understanding of spinal deconditioning during 15- and 30-days microgravity and of the higher incidence of disc prolapse or herniation following re-exposure to 1-G with a long-term view to prevent such spinal deconditioning with exercise or other physiologic countermeasures. This research may aid understanding of spinal deconditioning during inactivity such as after spinal cord injury and bed rest in human patients on Earth. Separately, our studies of mouse skull calvaria demonstrate the influence of the fluid shift associated with microgravity on bone remodeling independent of skeletal loading. The findings may be considered in the development of strategies to counter bone loss in astronauts and patients confined to bed rest.
Task Progress:	To test our hypotheses, we harvested and will test spines from ground-based control mice and flight mice exposed to 30-days of actual microgravity on Bion-M1. Spines were frozen and will be transported to UCSD and UCSF for accepted methods to test intervertebral disc morphology, cell content, water content, swelling pressure, GAG and PG concentrations, and spine biomechanics. We will compare the results of the present study to our peer-reviewed and published results for identical twins exposed to 30-days simulated microgravity, STS-131 flight mice, and control mice exposed to 1-G. Our research will aid understanding of spinal deconditioning during 30-days microgravity and of the higher incidence of disc prolapse or herniation following re-exposure to 1-G with a long-term view to prevent such spinal deconditioning with exercise or other physiologic countermeasures. When the specimens arrive at our facilities at UCSD and UCSF, we plan: (1) To measure intervertebral disc morphology, cell content, swelling pressure, water content, and GAG and PG concentrations after exposure to 30 days of microgravity and compare to STS-131 flight mice and ground-control animals. (2) To measure time-dependent and time-independent disc compression stiffness after exposure to 30 days of microgravity and compare to STS-131 flight mice and ground-control animals.
Bibliography Type:	Description: (Last Updated: 08/25/2025)
Articles in Peer-reviewed Journals	Mateus J, Hargens AR. "Photoplethysmography for non-invasive in vivo measurement of bone hemodynamics." Physiological Measurement. 2012 Jun;33(6):1027-42. http://dx.doi.org/10.1088/0967-3334/33/6/1027 ; PubMed PMID: 22562998 , Jun-2012
Articles in Peer-reviewed Journals	Sayson JV, Lotz J, Parazynski S, Hargens AR. "Back pain in space and post-flight spine injury: Mechanisms and countermeasure development." Acta Astronautica. 2013 May-Jun;86:24-38. http://dx.doi.org/10.1016/j.actaastro.2012.05.016 , May-2013
Articles in Peer-reviewed Journals	Goswami N, Batzel JJ, Clement G, Stein TP, Hargens AR, Sharp MK, Blaber AP, Roma PG, Hinghofer-Szalkay HG. "Maximizing information from space data resources: a case for expanding integration across research disciplines." European Journal of Applied Physiology. 2013 Jul;113(7):1645-54. Epub 2012 Oct 17. http://dx.doi.org/10.1007/s00421-012-2507-5 ; PubMed PMID: 23073848 (originally reported as Epub Oct 17, 2012) , Jul-2013
Articles in Peer-reviewed Journals	Mateus J, Hargens AR. "Bone hemodynamic responses to changes in external pressure." Bone. 2013 Feb;52(2):604-10. Epub 2012 Nov 17. http://dx.doi.org/10.1016/j.bone.2012.11.010 ; PubMed PMID: 23168293 , Feb-2013